

Notes: Smith (RES, 2004)

Supermarket Choice and Supermarket Competition in Market Equilibrium

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1 Big picture

- **Question.** How much market power do supermarket chains get from owning multiple nearby stores, and what would happen to prices under demerger or merger?
- **Setting.** The paper studies the U.K. supermarket industry, a highly concentrated retail market in which the large chains operate many stores and often compete locally with their own outlets.
- **Core empirical challenge.** Supermarket prices are hard to observe at the product-store level, while costs and product quality are also difficult to measure directly. The paper therefore reverses the usual strategy: instead of using prices to infer costs, it uses observed gross profit margins to identify the price sensitivity parameters in consumer demand.
- **Main conceptual contribution.** The paper builds a structural model that combines store choice, expenditure choice, and multi-store Nash pricing. The key mechanism is that a chain internalizes cannibalization across its own stores, which raises equilibrium prices relative to a world in which each store is independently owned.
- **Data advantage.** The paper combines three rich sources: observed firm-level gross profit margins, a complete store-level data set on supermarket characteristics, and a household shopping survey with more than 114,000 respondents.
- **Main empirical design.** Consumers choose a store and an expenditure level as a function of distance, store size, parking, firm identity, income, and shopping mode. Demand estimates are then fed into a Nash pricing system to recover equilibrium margins and run merger and demerger counterfactuals.
- **Main baseline result.** A complete demerger, where every store prices independently, reduces the revenue-weighted supermarket price index by about 1.7% on average. Prices of the largest firms fall by roughly 2% to 3.8% depending on local concentration.
- **Main merger result.** Hypothetical mergers among the big four increase prices, with Tesco mergers creating the largest effects. The Tesco–Sainsbury merger raises the revenue-weighted price index by about 1.5% on average and pushes the smaller merging firm’s prices up by more than 4% on average, with some regional maxima near 7.4%.
- **Policy message.** Multiple-store ownership generates economically meaningful market power. The paper therefore cautions against mergers among large chains and suggests that, in some highly concentrated regions, store reallocations or break-up remedies could lower prices and raise welfare.

2 Data and measurement (key objects)

2.1 Why the U.K. supermarket industry is a useful laboratory

- The U.K. supermarket sector is highly concentrated. In 1998, the largest four chains accounted for 71.2% of industry sales.
- The economic issue is not only concentration at the firm level, but *local overlap* across stores of the same chain. Standard oligopoly logic implies that if two stores owned by the same firm

are close substitutes, the firm will price both stores less aggressively because it internalizes lost sales at one store that are recaptured at another.

- This setting is therefore ideal for quantifying the market-power effect of multi-store ownership.

Interpretation. The paper is fundamentally about within-chain substitution. A merger matters because stores previously treated as rivals are now priced jointly; a demerger matters because stores previously coordinated are now priced independently.

2.2 Three main data sources

- **Gross profit margins by firm.** These are the paper’s key indirect source of price information.
- **Store characteristics.** The store database covers all 1,207 supermarket stores in the study area and records firm identity, location, floorspace, and parking availability.
- **Household shopping survey.** The National Shoppers Survey includes 114,058 households in January 1995. It records the firm used for primary shopping, the firms used for secondary shopping, expenditure categories, household income, and location.

Interpretation. The model is estimated from *who shops where* and *how much they spend*, while the pricing side is pinned down using profit margins rather than direct product-level prices.

2.3 Industry facts and descriptive objects

- The paper studies 11 supermarket firms in southern England. The big four are ASDA, Safeway, Sainsbury, and Tesco; the others are Budgen, Co-op, Iceland, Kwik Save, M&S, Somerfield, and Waitrose.
- Table 1 shows that the big four have larger stores, more product lines, and substantial market shares. Non-big-four firms tend to have smaller stores and more niche positioning.
- Table 2 shows strong heterogeneity in client mix: Sainsbury, Tesco, and Waitrose disproportionately attract higher-income primary shoppers, whereas Co-op, Somerfield, Kwik Save, and Iceland attract more low-income shoppers. Iceland and M&S are especially important for secondary shopping.
- Table 3 decomposes costs into cost of goods, labor and distribution, and other store costs, and reports gross profit margins that lie around 10%–14% for the major chains.

Interpretation. Firm identity carries a lot of economic content. It is not just a label: it proxies for quality, product range, freshness, service, and shopping format.

2.4 Consumers, markets, and types

- The unit of observation in demand estimation is a household shopping decision.
- Each consumer is characterized by income, location, and shopping mode.
- Income is grouped into three bands, shopping mode is either primary or secondary, and location is measured using 1,027 Postal Sectors.

- Combining these gives $3 \times 2 \times 1,027 = 6,162$ consumer types.
- Postal Sectors are grouped into nine pricing regions. Taste parameters vary by region, income, and mode, giving $9 \times 3 \times 2 = 54$ taste groups.

Interpretation. The paper combats the unrealistic substitution patterns of a plain multinomial logit by allowing taste parameters to vary across many narrowly defined consumer groups.

2.5 Choice set, outcomes, and main variables

Each consumer chooses from the nearest 30 stores plus a convenience-store outside option. The main store-level observables are

$$z_{ij} = (dist_{ij}, size_j, park_j, F_j),$$

where $dist_{ij}$ is distance from consumer location i to store j , $size_j$ is floorspace, $park_j$ indicates car parking, and F_j is the operating firm.

For each household, the survey also records expenditure categories rather than exact spending amounts. This is important because the model jointly explains:

- the discrete choice of *which firm/store to visit*, and
- the continuous choice of *how much to spend conditional on visiting*.

Interpretation. This discrete-plus-continuous structure lets the model distinguish between stores that attract many customers and stores that attract high-spending customers.

2.6 Primary versus secondary shopping

- Primary shopping is the main weekly grocery trip.
- Secondary shopping is top-up or supplementary shopping and accounts for roughly 20% of total expenditure.
- The model treats the two decisions separately, allowing parameters to differ by mode.

Interpretation. This is economically useful because different firms specialize in different shopping occasions. For example, some firms attract large-basket weekly trips, while others are better suited to top-up visits.

3 Models and empirical specifications (all equations + interpretation)

3.1 Indirect utility and store choice

Consumer I of type i values store j according to

$$V_j^I = v(z_{ij}, p_j, \zeta^I; \theta_{g(i)}) + \epsilon_j^I = v_j^I + \epsilon_j^I. \quad (1)$$

Here:

- p_j is the price of a unit of groceries at store j ,
- ζ^I is an unobserved household characteristic, interpreted naturally as household size,
- ϵ_j^I is an idiosyncratic taste shock,
- $\theta_{g(i)} = (\alpha_g, \beta_g, \phi_g, \lambda_g, \mu_g)$ are group-specific parameters.

Conditional on ζ^I , store choice follows a multinomial logit. Integrating over the distribution of ζ^I gives the market share of store j among consumer type i :

$$r_{ij}(z_i, p; \theta_{g(i)}) = \int s_{ij}(z_i, p, \zeta^I; \theta_{g(i)}) dH(\zeta^I). \quad (2)$$

Interpretation. This is a random-coefficients style idea implemented in a very tractable way. Consumers differ in unobserved intensity of grocery demand, and the model integrates those differences out.

3.2 Utility functional form and conditional expenditure

The paper uses the following indirect utility specification:

$$V_j^I = \mu_g [\beta_2^g y_i + \beta_1^g \ln p_j + \phi_{1F}^g + \alpha_1^g \ln size_j] \exp((\phi_{2F}^g - \beta_2^g \ln p_j) \zeta^I) + \alpha_2^g park_j - \alpha_3^g \ln dist_{ij} + \epsilon_j^I. \quad (4)$$

Roy's identity implies conditional grocery demand

$$x_j^I = \frac{1}{p_j} \left[(\beta_2^g y_i + \beta_1^g \ln p_j + \phi_{1F}^g + \alpha_1^g \ln size_j) \zeta^I - \frac{\beta_1^g}{\beta_2^g} \right], \quad (5)$$

and conditional expenditure

$$e_j^I = (\beta_2^g y_i + \beta_1^g \ln p_j + \phi_{1F}^g + \alpha_1^g \ln size_j) \zeta^I - \frac{\beta_1^g}{\beta_2^g}. \quad (6)$$

The conditional price elasticity of demand is

$$\eta^I = -1 + \zeta^I \beta_1^g / (x_j^I p_j). \quad (7)$$

Interpretation. The important economic innovation is that high-volume stores face not only different market shares but also different spending per customer. This matters because stores that induce larger baskets also make consumers more price sensitive in absolute terms.

3.3 From store-level choice to observed firm-level data

Expenditure is observed in categories, not in exact levels. Let q_{ijl} be the probability that consumer type i shops at store j and falls into expenditure category l :

$$q_{ijl} = \int_{\zeta_l}^{\zeta_{l+1}} s_{ij}(z_i, p, \zeta^I; \theta_{g(i)}) dH(\zeta^I). \quad (9)$$

Aggregating stores to the firm level gives the probability of choosing firm F with expenditure category l :

$$Pr(e_i = l, F_i = F \mid z_i, p; \theta_{g(i)}) = \sum_{\{j: F_j = F\}} \int_{\zeta_l}^{\zeta_{l+1}} s_{ij}(z_i, p, \zeta^I; \theta_{g(i)}) dH(\zeta^I). \quad (10)$$

The likelihood for group g is then

$$\ln L_g = \sum_{i \in g} \sum_l \sum_F D_{lF}^i \ln Pr(e_i = l, F_i = F \mid z_i, p; \theta_{g(i)}). \quad (11)$$

Interpretation. The survey does not identify the exact store, only the firm used. The model therefore predicts store-level probabilities and then aggregates them up to match the observable firm-level shopping outcome.

3.4 Multi-store Nash pricing and profit margins

Let firm F own a set of stores J_F and set a common price p_F over those stores in a pricing region. Conditional on a consumer type, gross profit is

$$\Pi_F^i = M_i \sum_{j \in J_F} \delta_F s_j^i x_j^i, \quad (12)$$

where $\delta_F = p_F - c_F$ is the markup.

The first-order condition for price, before integrating over consumer heterogeneity, is

$$\frac{\partial \Pi_F^i}{\partial p_F} = M_i \sum_{j \in J_F} s_j^i \left[\delta_F x_j^i (1 - s_j^i) \frac{\partial v_j^i}{\partial p_F} + x_j^i - \sum_{k \in J_F} \delta_F x_k^i s_k^i \frac{\partial v_j^i}{\partial p_F} + \delta_F \frac{\partial x_j^i}{\partial p_F} \right] = 0. \quad (13)$$

After summing across consumer types and integrating over ζ^I , the pricing equation can be written compactly as

$$m_F = m_F(M, z, p; \theta). \quad (17)$$

Interpretation.

- The standard single-store logit pricing terms are still present.
- The new and central term is the sum over $k \in J_F$: when the firm owns many stores, it internalizes demand that switches across its own outlets.
- That is exactly the source of multi-store market power studied in the paper.

3.5 Estimation strategy: two stages

The paper estimates the model in two stages.

Stage 1: demand and expenditure.

- Prices are not directly observed at the relevant pricing-region level, so they are absorbed into transformed firm parameters:

$$d_{1F}^g \equiv \beta_2^g y_i + w_g \beta_1 \ln p_F + \phi_{1F}^g, \quad (18)$$

$$d_{2F}^g \equiv \exp(\phi_{2F}^g - \beta_2 \ln p_F). \quad (19)$$

- The transformed utility and expenditure equations are then estimated by maximum likelihood separately for each taste group.

Stage 2: pricing.

- Using the stage-1 estimates, the paper estimates the price-sensitivity parameter β_2 by nonlinear least squares, matching model-implied profit margins to observed gross profit margins:

$$\bar{m}_F = m_F(M, z; \hat{\theta}, \beta_2) + \varepsilon_F. \quad (25)$$

- The objective function is

$$\sum_F (\bar{m}_F - m_F(M, z; \hat{\theta}, \beta_2))^2. \quad (26)$$

Interpretation. The paper’s identification logic is elegant: shopping data identify the nonprice side of demand, and margin data identify the price side.

3.6 Preferred specification and model fit

The paper considers four specifications that differ in the assumed ratio β_1/β_2 , which changes the implied conditional demand elasticity. Specification III is preferred because it matches the benchmark elasticity target best.

From Table 9:

- estimated $\beta_2 = 0.970$,
- implied $\beta_1 = 9.700$,
- implied conditional elasticity for a median consumer is about -0.689 ,

which is very close to the target value of about -0.7 based on external U.K. demand estimates.

The model’s fit is checked in several ways:

- Table 10 shows plausible shopping distances;
- Table 11 shows substitution patterns that vary sensibly by income and shopping mode;
- Table 12 shows predicted gross margins that line up closely with observed firm margins.

Let t index an ownership structure. A demerger or merger changes which stores belong to which firm, which changes the Nash margins m_t and prices p_t .

The counterfactual pricing system is solved from

$$m_t = m(M, z_t; \theta_0, d_t), \quad (27)$$

with the transformed firm parameters updated as prices change:

$$d_{1j}^t = d_{1j}^0 + w_g \beta_1^0 (\ln p_j^t - \ln p_j^0), \quad (28)$$

$$d_{2j}^t = d_{2j}^0 \exp(-\beta_2^0 (\ln p_j^t - \ln p_j^0)). \quad (29)$$

The system is then solved numerically. Consumer surplus is computed from the log-sum formula,

$$CS^I = \sigma^I \ln \sum_{j \in J_i} \exp(v_j^I). \quad (31)$$

Interpretation. The counterfactuals are clean but partial equilibrium in one important sense: marginal costs and nonprice attributes are held fixed. So the experiments isolate the pricing effect of ownership structure, not long-run changes in store quality or investment.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the structural parameters

The paper has no external IV in the usual reduced-form sense. Identification comes from model structure plus multiple observables.

- ϕ_{1F}^g is identified from variation in expenditure across firms within a taste group.
- ϕ_{2F}^g is identified from variation in market shares relative to expenditure.
- α_g on observable store characteristics is identified because firm dummies vary at the firm level while observables such as size, parking, and distance vary at the store level.
- λ_g is identified from the spread of expenditure across categories.
- β_2 is identified in stage 2 by matching predicted and observed profit margins.

Interpretation. The paper uses a rich mapping from observables to primitives: choice identifies attractiveness, expenditure identifies intensity of demand, and margins identify price sensitivity.

4.2 Why the absence of store-specific unobservables is plausible here

A typical differentiated-products problem is that product dummies absorb most variation in observed attributes. Smith argues that this is less severe in the supermarket context because the most important hard-to-measure characteristics are chosen at the *firm* level rather than the store level: layout, service speed, freshness, decor, and product line quality are chain-wide decisions.

Interpretation. That assumption is not innocuous, but in supermarkets it is more defensible than in many other retail settings because brand identity and chain-wide standards are unusually strong.

4.3 How price is handled despite being unobserved

Because pricing-region prices are not observed directly, the paper absorbs price into transformed firm parameters in stage 1 and then recovers the price-side parameter in stage 2 from gross margins. The convenience-store outside option is normalized by setting

$$d_{2F0}^g = 1,$$

which anchors utility for the outside option.

The paper also imposes two simplifying restrictions:

- $\beta_2^g = \beta_2$ for all groups, so price sensitivity of utility is common across groups;
- $\beta_1^g = w_g \beta_1$, so conditional-demand price elasticity scales with average expenditure by group.

Interpretation. These restrictions are driven by data limitations: there are only as many profit-margin observations as firms, so fully group-specific price parameters cannot be identified.

4.4 Why simultaneity is limited in stage 2

The paper argues that once the pricing equation is rewritten as

$$m_F = m_F(M, z; \hat{\theta}, \beta_2),$$

prices no longer enter as direct arguments in the second-stage regression. Taking store characteristics z as predetermined, the nonlinear least-squares problem for β_2 is therefore not subject to the standard demand-estimation simultaneity issue.

Interpretation. This is a structural version of “using a different moment to identify price sensitivity.” The pricing side is disciplined by margins rather than by direct price variation.

4.5 Main maintained assumptions and limitations

- Firms set uniform prices within nine pricing regions, and four firms are assumed to price uniformly nationally.
- The observed gross profit margins are national while the model is estimated on a subnational environment; the paper treats the gap partly as measurement error.
- The pricing game is static multi-store Nash, not dynamic pricing and not tacit collusion.
- In merger and demerger experiments, marginal costs and store quality characteristics are held fixed.

Interpretation. The counterfactuals are best read as short-run pricing effects under a fixed store network and fixed cost/quality primitives.

4.6 Relation to the literature

The paper sits at the intersection of several literatures:

TABLE 1
The supermarkets

	Market share ^c (%)	Avg lines ^d	Own label (%)	Avg size ^e (sq ft)	Price ^b Branded basket (£)	Storewise range ^c	
						All prices (%)	Varying prices (%)
Big four:							
ASDA	13.4	38	34	45.2	25.89	0	0
Safeway	12.5	22	41	22.1	27.23	1.1	4.3
Sainsbury	20.7	21	54	29.4	27.16	NA ^f	NA ^f
Tesco	24.6	31	45	27.0	27.21	1.7	19.2
Others:							
Budgen	0.4	9	27	8.4	NA	3.1	9.8
Co-op	4.2	10	27	13.3	28.7	0.5	6.7
Iceland	0.1	3	50	5.8	25.80	0	0
Kwik Save	NA ^f	6	13	9.0	25.39	0.8	9.8
M&S	5	8	100	6.4	—	0	0
Somerfield	8.5	12	33	8.5	27.52	0.2	6.3
Waitrose	3.3	13	39	13.8	28.47	0	0

Source: ^a Institute of Grocery Distribution; ^b Consumer Assoc; ^c CC (2000); ^d Units in 1000s; ^e Sainsbury varies prices but provided no data; ^f Kwik Save was acquired by Somerfield in 1997.

supermarket firms in southern England. Their characteristics for the U.K. market as a whole are summarized in Table 1. The big four are ASDA, Safeway, Sainsbury and Tesco. Compared

- differentiated-products demand and equilibrium pricing (BLP, Goldberg, Nevo),
- spatial competition,
- and supermarket competition and markups.

Its distinctive contribution is to use gross profit margins instead of prices to estimate the price side of demand in a multi-product, multi-location retail setting where collecting a full price panel is extremely difficult.

5 Tables and figures

5.1 A note on presentation

This article is almost entirely **table-driven**. It does not rely on standalone figures in the way many modern empirical papers do. The evidence is carried by descriptive tables, parameter tables, model-fit tables, and merger/demerger counterfactual tables.

5.2 Table 1: the supermarkets

Read.

- The big four dominate market shares.
- They tend to operate larger stores and wider product ranges.
- Some firms vary prices across stores or regions; others do not.

Economic message. This table motivates the basic problem: a concentrated market with many multi-store chains and potential within-chain local substitution.

5.3 Tables 2 and 3: who shops where, and what margins look like

TABLE 2
Primary and secondary: shoppers and market shares

	Primary shoppers Household income ^a				Secondary shoppers Household income ^a			
	Low	Medium	High	All	Low	Medium	High	All
Market shares (%):								
ASDA	10.4	11.5	9.5	10.1	6.3	7.4	7.4	7.0
Safeway	9.2	10.1	9.8	9.6	7.9	7.8	5.7	8.3
Sainsbury	17.5	20.9	27.2	21.4	11.4	13.3	15.2	13.5
Tesco	22.3	29.5	33.6	27.8	13.8	16.0	17.5	15.6
Budgen	1.1	0.8	0.6	0.9	0.5	0.7	1.1	0.7
Co-op	9.3	6.2	3.5	6.7	9	7.8	5.8	7.6
Iceland	1.3	0.6	0.3	0.8	9.1	7.8	5.7	7.7
Kwik Save	5.8	3.4	1.1	3.8	6.2	3.4	2.7	4.5
M&S	0.6	0.5	0.6	0.5	7.9	7.8	10.5	8.7
Somerfield	14.4	10.4	6.3	10.9	13.7	12.1	8.8	11.7
Waitrose	2.3	2.3	5.1	3.1	2.3	2.8	5.7	3.5
Corner/Other	5.7	3.7	2.3	4.1	11.9	11.1	11.9	11.1

Source: National Shoppers Survey. ^a Low < £10k; medium £10k – 20k; high > £20k.

TABLE 3
Costs and profits as percentage of sales

	Store and distribution costs					Profit margin		
	Variable costs					(i)	(ii)	(iii)
	Cost of goods	Dist'n & labour	Total v'ble	Other	Total store			
	(1)	(2)	(3)	(4)	(5)	100–(1)	100–(3)	100–(5)
ASDA	77.5	10.6	88.1	3.5	91.6	22.5	11.9	9.4
Safeway	—	11.5	88.6	4.8	93.4	—	11.4	6.6
Sainsbury	—	11.7	86.0	4.2	90.2	—	14.0	9.8
Tesco	—	11.8	87.1	4.9	92.0	—	12.9	8.0
Budgen	90.5 ^a	5.5	96.0	—	—	—	4.0	—
Iceland	—	—	—	—	91.7	—	—	8.3
Kwik Save	85.3	9.9	95.2	—	—	14.7	4.8	—
M&S	72.0	14.4	86.4	—	—	28.0	13.6	—
Somerfield	77.9	10.3	88.4	5.3	93.7	22.1	11.6	6.3
Waitrose	77.1	12.5	89.6	—	—	22.9	10.4	—

Source: CC (2000), Company accounts, HSBC James Capel (1996). ^a Incl. labour; (3) = (1) + (2); (5) = (4) + (3).

Read.

- Table 2 shows systematic variation in shopper composition by income and shopping mode.
- Table 3 shows that gross margins are materially positive and vary across firms.

Economic message. Supermarkets are differentiated products, not homogeneous grocery boxes. Consumer segmentation matters, and there is room for economically meaningful markup variation.

5.4 Tables 4 and 5: sample structure and regional variation

TABLE 4
Descriptive statistics

Store data: Variable	Mean	S.D.	Min	Max
Size (sq ft)	16,250.7	14,233.8	616.0	105,600.0
Park	0.6	0.5	0.0	1.0
Dist (km) ^a	14.7	12.0	0.0	138.0
# Stores	1207			

Source: IGD, Goad, ^a consumer/store pairs in choice sets.
S.D. = standard deviation.

Consumer data:		
#Respondents ^a	Low income	47,167
	Medium income	33,246
	High income	33,645
	Total:	114,058
Sample as % of all households		4.8%
#Postal sectors in area		1027
Mean # households (hh) per postal sector		2309
Standard deviation # hh per postal sector		(1190)

Source: National Shoppers Survey; ^a low < £10k; medium £10k–20k; high > £20k.

TABLE 5
Pricing regions: demographics and distribution of stores

	BATA	BHDT	BS	EXTQ	OX	PLTR	RG	SNSP	SO	All
Households and population (units in 100,000s):										
Households	2.9	3.0	2.8	3.2	2.0	3.6	2.1	2.0	2.3	23.7
Population	6.9	7.5	6.8	8.0	5.1	9.3	5.1	5.1	5.6	59.4
% in household income group: ^a										
Low	43	46	43	48	31	49	27	37	41	41
Medium	31	30	29	32	27	32	23	30	26	29
High	26	24	27	20	42	19	49	33	34	29
Number of stores by firm:										
ASDA	2	3	4	0	1	2	2	2	4	20
Safeway	7	10	3	5	0	7	3	4	6	45
Sainsbury	5	6	5	5	6	3	7	5	6	48
Tesco	4	11	8	7	7	10	6	6	6	65
Budgen	1	0	1	0	4	0	3	3	1	13
Co-op	6	11	11	14	13	17	4	5	5	86
Iceland	2	3	6	2	2	3	4	6	3	31
Kwik Save	9	4	9	2	3	2	1	5	3	38
M&S	4	5	3	5	2	3	3	3	3	31
Somerfield	18	15	21	34	9	15	10	15	2	139
Waitrose	1	4	1	0	5	0	6	3	3	23
Floorspace concentration:										
C2	31	39	38	43	45	40	52	35	41	37

Source: National Shoppers Survey, IGD, Goad; ^a low < £10k; medium £10–20k; high > £20k.

Read.

- Table 4 documents the scale of the store and consumer data.
- Table 5 shows strong regional differences in demographics, store presence, and concentration.

Economic message. The paper is estimating one structural framework over a rich and heterogeneous geographic environment. Regional concentration and consumer mix are central for the counterfactuals.

5.5 Tables 6–10: parameter estimates and plausibility checks

TABLE 6
Primary shopping parameters (specification I)

Parameter	Variable	Low income			Medium income			High income		
		Estimate	S.E.	S.D.	Estimate	S.E.	S.D.	Estimate	S.E.	S.D.
d_1	ASDA	25.144	2.542	0.205	31.785	4.180	10.690	24.202	6.871	8.133
	Sainsbury	22.073	2.263	0.133	29.093	3.320	11.381	25.281	5.245	10.261
	Waitrose	22.193	2.209	0.273	27.365	3.656	12.345	27.266	6.036	7.366
	Budget	24.095	2.075	0.231	28.976	3.656	12.321	24.644	5.663	10.903
	Co-op	24.044	2.837	0.099	35.837	4.761	5.165	35.454	5.533	7.180
	Waitrose	21.742	1.715	0.054	29.275	3.726	9.606	26.849	4.893	9.513
	M&S	26.568	2.660	0.071	32.295	4.651	9.294	30.321	6.806	7.829
	Kwik Save	21.880	1.597	0.211	27.940	2.967	8.744	21.879	4.709	7.455
	Somerfield	22.530	3.374	0.105	31.581	5.465	8.893	40.788	7.136	8.103
	Waitrose	22.185	1.550	0.136	29.424	3.029	8.002	28.929	4.087	8.250
	Waitrose	22.097	2.188	0.218	33.737	3.864	7.469	33.781	4.967	6.886
	Waitrose	24.089	0.847	1.365	36.872	1.610	2.099	45.582	2.333	2.968
	Waitrose	1.001	0.148	0.205	1.317	1.610	0.054	1.729	0.476	0.768
	Waitrose	1.047	0.102	0.133	1.285	0.161	0.407	1.647	0.392	0.625
	Waitrose	1.123	0.140	0.273	1.434	0.225	0.512	1.851	0.448	0.712
d_2	ASDA	1.052	0.129	0.231	1.409	0.230	0.518	1.867	0.491	0.727
	Budget	0.930	0.094	0.099	0.951	0.109	0.061	1.016	0.126	0.113
	Co-op	1.051	0.086	0.054	1.144	0.125	0.229	1.226	0.152	0.279
	Waitrose	0.810	0.079	0.071	0.955	0.135	0.152	0.942	0.279	0.340
	Kwik Save	1.145	0.123	0.211	1.332	0.167	0.343	1.503	0.235	0.296
	M&S	0.823	0.124	0.195	0.875	0.160	0.244	0.882	0.194	0.195
	Somerfield	1.069	0.088	0.136	1.231	0.146	0.263	1.357	0.193	0.379
	Waitrose	1.059	0.104	0.218	1.137	0.158	0.277	1.450	0.312	0.564
	Size	1.654	0.614	1.697	3.504	1.011	1.011	7.849	1.603	3.038
	Park	0.249	0.094	0.437	0.416	0.142	0.142	0.604	0.155	0.522
	Dist	1.075	0.022	0.139	1.069	0.027	0.111	1.139	0.029	0.100
	Scale	2.665	0.032	0.067	3.095	0.043	0.142	3.227	0.048	0.114
	μ	0.596	0.139	0.602	0.617	0.179	1.122	0.134	0.050	0.089
	LLF	-147,449.27			-105,509.23	0.031	0.031	-103,529.25	0.035	0.089
	OBS	47,167			33,246			33,645		

Estimate: average value of regional parameters.
S.E.: average value of regional standard errors.
S.D.: standard deviation of regional parameters.
LLF: sum of regional likelihood values.
OBS: sum of regional number of observations.

TABLE 7
Secondary shopping parameters (specification I)

Parameter	Variable	Low income			Medium income			High income		
		Estimate	S.E.	S.D.	Estimate	S.E.	S.D.	Estimate	S.E.	S.D.
d_1	ASDA	6.040	0.364	2.635	6.820	0.034	2.695	7.631	0.403	3.204
	Sainsbury	6.574	0.300	2.608	7.067	0.036	2.752	8.144	0.299	3.746
	Waitrose	6.809	0.279	1.124	7.574	0.034	0.866	8.589	0.269	1.686
	Budget	6.863	0.270	1.143	7.599	0.029	0.770	8.485	0.251	1.620
	Co-op	7.456	0.306	0.850	8.470	0.059	0.590	9.849	0.727	1.068
	Waitrose	7.846	0.349	0.769	9.199	0.062	0.456	10.523	0.732	1.008
	Kwik Save	7.518	0.353	0.892	8.934	0.057	0.508	9.690	1.377	1.104
	M&S	7.663	0.392	0.891	9.085	0.069	0.577	10.611	0.559	1.259
	Somerfield	7.528	0.225	0.840	8.925	0.043	0.581	10.523	1.042	0.712
	Waitrose	5.477	0.318	3.185	6.715	0.046	3.856	7.613	0.397	4.488
	Waitrose	7.971	0.241	0.402	9.233	0.274	0.337	10.821	0.293	0.593
	Waitrose	0.979	0.044	0.420	0.950	0.306	0.362	0.981	0.046	0.371
	Waitrose	0.914	0.042	0.372	0.885	0.308	0.339	0.926	0.030	0.350
	Waitrose	1.102	0.032	0.181	1.098	0.281	0.072	1.159	0.026	0.076
	Waitrose	1.086	0.032	0.140	1.089	0.254	0.053	1.161	0.025	0.087
d_2	ASDA	0.605	0.070	0.409	0.742	0.480	0.577	0.725	0.066	0.364
	Budget	0.924	0.041	0.087	0.893	0.385	0.104	0.866	0.086	0.154
	Co-op	1.044	0.066	0.136	0.977	0.432	0.066	0.973	0.049	0.051
	Kwik Save	0.980	0.060	0.071	0.902	0.442	0.045	0.911	0.140	0.163
	M&S	1.074	0.071	0.121	1.002	0.441	0.031	1.060	0.070	0.092
	Somerfield	0.944	0.038	0.136	0.877	0.319	0.078	0.886	0.126	0.101
	Waitrose	0.754	0.047	0.440	0.728	0.360	0.416	0.769	0.038	0.441
	Size	0.340	0.078	0.299	0.433	0.074	0.264	0.542	0.063	0.378
	Park	0.129	0.053	0.323	0.304	0.082	0.328	0.322	0.083	0.309
	Dist	0.733	0.017	0.098	0.707	0.018	0.095	0.705	0.019	0.102
	Scale	1.746	0.029	0.068	1.842	0.030	0.065	1.702	0.024	0.056
	μ	2.436	0.069	3.830	0.655	0.222	0.638	0.893	0.061	0.850
	LLF	-140,011.49			-101,502.64			-106,668.38		
	OBS	47,167			33,246			33,645		

Estimate: average value of regional parameters.
S.E.: average value of regional standard errors.
S.D.: standard deviation of regional parameters.
LLF: sum of regional likelihood values.
OBS: sum of regional number of observations.

TABLE 8
Primary and secondary parameters (specification III)

Parameter	Variable	Low income			Medium income			High income		
		Estimate	S.E.	S.D.	Estimate	S.E.	S.D.	Estimate	S.E.	S.D.
Primary	Size	1.666	0.544	1.649	3.123	1.035	2.816	6.982	1.684	2.978
	Park	0.249	0.103	0.435	0.416	0.142	0.434	0.604	0.155	0.526
	Dist	1.075	0.022	0.139	1.069	0.027	0.111	1.139	0.029	0.100
	Scale	2.665	0.032	0.067	3.095	0.043	0.142	3.227	0.048	0.114
	μ	0.596	0.139	0.602	0.617	0.179	1.122	0.134	0.050	0.089
	LLF	-147,126.06			-104,892.21			-103,053.87		
Secondary	Size	0.363	0.080	0.330	0.426	0.250	0.074	0.455	0.055	0.400
	Park	0.130	0.057	0.323	0.305	0.330	0.092	0.322	0.075	0.391
	Dist	0.733	0.017	0.098	0.707	0.095	0.019	0.705	0.018	0.102
	Scale	2.114	0.035	0.082	2.329	0.080	0.037	2.174	0.030	0.072
	μ	0.703	0.056	0.469	0.794	0.532	0.235	1.614	0.282	1.909
	LLF	-140,011.47			-101,503.62			-106,470.45		

Note: Parameters on firm dummies not reported.
Estimate: average value of regional parameters.
S.E.: average value of regional standard errors.
S.D.: standard deviation of regional parameters.
LLF: sum of regional likelihood values.

TABLE 9
Price parameters

	Specification			
	I	II	III	IV
β_1	0	5	10	15
Estimated β_2	0.930	0.947	0.970	0.983
Standard error	(0.071)	(0.076)	(0.076)	(0.077)
Implied β_1	0.000	4.735	9.700	14.745
η^I (for $\zeta = 1$)	-1.000	-0.848	-0.689	-0.526

Observations = 9.

TABLE 10
Customers by distance to store (specification III)

Mode	Income	Proportion of customers by distance				
		<5 km	<10 km	<15 km	<20 km	<25 km
Primary	Low	0.415	0.599	0.756	0.903	0.973
	Medium	0.413	0.606	0.762	0.910	0.976
	High	0.424	0.620	0.772	0.916	0.977
Secondary	Low	0.404	0.579	0.735	0.891	0.969
	Medium	0.398	0.576	0.735	0.892	0.970
	High	0.401	0.579	0.737	0.895	0.972

Read.

- The preferred model is specification III because it produces a conditional elasticity near the external benchmark.
- Floorspace enters positively, distance enters negatively, and parking improves store attractiveness.
- Predicted shopping distances are plausible.

Economic message. The model is not only internally estimable; it also generates patterns that look reasonable when compared with outside evidence and common sense.

5.6 Tables 11 and 12: substitution patterns and model fit

Read.

- When Tesco raises price, high-income consumers substitute disproportionately toward Sainsbury and Waitrose.

- Secondary shoppers substitute more toward firms such as Iceland, M&S, and Corner/Other.
- Predicted margins line up closely with observed margins, especially for the major chains.

Economic message. The model captures the right substitution patterns across consumer segments and replicates the observed cross-firm margin structure sufficiently well to be useful for counterfactual policy work.

TABLE 11
Substitution by consumer type (specification III)

For a given price increase at Tesco % who substitute to each firm (as a % of total substituting from Tesco)						
Primary			Secondary			
	Low (%)	Medium (%)	High (%)	Low (%)	Medium (%)	High (%)
ASDA	15.05	18.05	15.55	8.70	10.35	9.82
Safeway	12.86	15.38	15.92	9.82	11.09	11.54
Sainsbury	23.70	30.60	40.81	14.83	18.08	22.83
Budgen	1.44	1.36	1.44	1.02	1.30	2.04
Co-op	12.07	8.86	5.32	10.71	9.20	6.83
Iceland	1.75	0.98	0.62	9.03	7.94	5.81
Kwik Save	8.10	5.18	1.92	7.16	5.16	3.07
M&S	0.79	0.67	1.01	8.97	9.02	12.12
Somerfield	16.83	13.79	9.09	15.78	14.15	10.48
Waitrose	4.35	5.09	11.23	3.57	4.53	9.14
Corner/Other	7.76	5.47	3.73	13.72	13.15	11.63

TABLE 12
Observed and predicted profit margins

	Observed	Predicted			
		I	II	III	IV
ASDA	0.119	0.110	0.115	0.115	0.116
Safeway	0.114	0.112	0.115	0.115	0.116
Sainsbury	0.140	0.114	0.119	0.121	0.123
Tesco	0.129	0.120	0.126	0.127	0.130
Budgen	0.040	0.133	0.127	0.123	0.121
Co-op	NA	0.137	0.135	0.134	0.133
Iceland	NA	0.142	0.135	0.135	0.132
Kwik Save	0.048	0.130	0.130	0.129	0.130
M&S	0.136	0.133	0.126	0.127	0.124
Somerfield	0.116	0.137	0.136	0.135	0.135
Waitrose	0.104	0.109	0.109	0.108	0.107

Observed figures as Table 3 column (ii).

ranges are slightly higher than those found by the CC (see Table 1) but in both cases the predicted

TABLE 14
Equilibrium effect of demerger (specification III)

	Pricing regions									
	BATA	BHDT	BS	EXTQ	OX	PLTR	RG	SNSP	SO	Avg
HHI	1267	1547	1544	1684	1970	1429	1827	1587	1686	1616
Δ HHI	-1016	-1312	-1214	-1423	-1596	-1129	-1411	-1320	-1311	-1303
Share 1	20.96	28.17	27.94	23.87	33.65	24.93	30.36	28.23	26.93	27.23
Share 2	15.64	17.67	17.37	20.61	23.95	14.43	25.16	21.99	22.87	19.96
% Δ wp	-1.13	-1.79	-1.55	-1.78	-2.18	-1.54	-1.46	-1.67	-1.90	-1.67
% Δ p ₁	-2.03	-3.17	-2.89	-3.76	-3.74	-3.48	-2.09	-2.79	-3.21	-3.02
% Δ p ₂	-1.04	-2.04	-1.89	-2.04	-2.69	-1.14	-2.26	-2.20	-2.13	-1.94
% Δ p ₃	-1.54	-1.06	-0.89	-2.09	-0.96	-1.51	-0.66	-1.57	-1.60	-1.32
% Δ p ₄	-6.90	-9.92	-8.22	-8.91	-12.69	-6.59	-10.07	-10.14	-10.91	-9.37
% Δ p ₅	-6.88	-15.10	-13.66	-13.95	-17.71	-12.68	-9.00	-13.12	-14.45	-12.95
% Δ p ₆	-5.94	-13.80	-11.52	-14.01	-22.85	-9.82	-14.77	-12.04	-15.62	-13.37
% Δ w/R	-0.98	-1.54	-1.36	-1.52	-1.91	-1.33	-1.28	-1.48	-1.67	-1.45
% Δ CS/R	2.01	2.92	2.62	3.77	2.92	2.43	1.64	2.49	2.92	2.62
% Δ W/R	1.03	1.38	1.26	2.25	1.01	1.10	0.36	1.01	1.25	1.17

Note: Share i , p_i , π_i are share, price, and gross profits of i -th largest firm, respectively.

$$CS^I = \sigma^I \mathcal{E}(\max_{j \in J^I} V_j^I) = \sigma^I \ln \sum_{j \in J^I} \exp(v_j^I) \quad (31)$$

where σ^I converts utility into monetary terms and is given by the inverse of the marginal utility of income (this depends on chosen firm F , so we use a market share weighted average for each consumer i -type). Total consumer surplus CS is obtained by integrating out over the distribution for ζ^I , multiplying expected utility for each i -type by the number of consumers of that type, and summing over the entire set of i -types.

7.1. Demerger

5.7 Tables 14–16: demerger, merger, and robustness

Read.

- Table 14 shows that demerger sharply reduces concentration and lowers prices, especially for the largest firms in the most concentrated regions.
- Average revenue-weighted prices fall by 1.67%; the largest-firm price falls by 3.02% on average; consumer surplus rises by 2.62% of initial industry revenue; total welfare rises by 1.17%.
- Table 15 shows that mergers raise prices, especially when Tesco is one of the merging firms. The Tesco–Sainsbury merger is the most important case, with a 1.48% increase in the overall price index and a 4.44% increase in the smaller merging firm’s price on average.
- Table 16 shows the demerger and merger conclusions are robust across alternative elasticity assumptions.

Economic message. The model’s policy conclusions are economically large and not fragile. Multi-store pricing power matters enough to affect prices, profits, consumer surplus, and welfare in a meaningful way.

TABLE 15
Equilibrium effect of merger (specification III)

	Tesco			Sainsbury		Safeway	Somerfield
	ASDA	Safeway	Sain	ASDA	Safeway	ASDA	Kwik Save
HHI	1607	1571	1616	1607	1571	1555	1575
ΔHHI	346	325	528	297	296	194	82
Share 1	25.18	23.50	22.66	18.91	18.63	10.28	11.93
Share 2	7.90	7.72	16.06	8.38	8.36	9.23	3.21
% Δwp	0.65	0.64	1.48	0.55	0.48	0.29	0.17
	(1.31)	(0.95)	(2.73)	(1.08)	(0.72)	(0.56)	(0.26)
% Δp_1	1.29	1.35	3.20	1.39	1.18	1.21	0.64
	(2.66)	(2.13)	(5.00)	(2.40)	(1.81)	(1.68)	(1.15)
% Δp_2	3.77	3.95	4.44	2.95	2.49	1.26	2.34
	(4.99)	(6.55)	(7.39)	(4.18)	(3.72)	(2.44)	(4.89)
% Δp_0	0.08	0.08	0.07	0.09	0.09	0.06	0.04
% $\Delta \pi_1$	1.96	2.04	3.46	1.53	1.20	0.40	0.45
% $\Delta \pi_2$	-0.10	-0.22	1.32	0.12	0.09	0.37	-0.15
% $\Delta \pi_0$	6.17	6.02	14.97	5.05	4.45	2.49	1.30
% $\Delta \pi/R$	0.56	0.56	1.19	0.49	0.43	0.27	0.14
% $\Delta CS/R$	-0.85	-1.09	-1.76	-0.73	-0.79	-0.49	-0.27
% $\Delta W/R$	-0.29	-0.53	-0.57	-0.24	-0.36	-0.22	-0.13

Note: Share 1, p_1 , and π_1 are market share, price, and gross profits of larger merging firm, p_0 and π_0 are for non-merging firms. Maximum regional values in parentheses.

concentrated region, the figure is highest: 2.69%. In BATA, the least concentrated, the figure is lowest: 1.12%.

TABLE 16

Demerger and merger: alternative specifications

	Demerger				Sainsbury and Tesco mergers				
	under alternative specifications:				under alternative specifications:				
	I	II	III	IV	I	II	III	IV	
% Δwp	-1.56	-1.63	-1.67	-1.72	% Δwp	1.35	1.44	1.48	1.52
% Δp_1	-2.76	-2.92	-3.02	-3.13	% Δp_1	2.93	3.10	3.20	3.29
% Δp_2	-1.77	-1.88	-1.94	-2.01	% Δp_2	4.07	4.30	4.44	4.57
% Δp_3	-1.16	-1.29	-1.32	-1.36	% Δp_0	0.04	0.07	0.07	0.09
% $\Delta \pi_1$	-9.50	-9.39	-9.37	-9.29	% $\Delta \pi_1$	3.80	3.67	3.46	3.30
% $\Delta \pi_2$	-12.51	-12.65	-12.95	-13.20	% $\Delta \pi_2$	1.70	1.55	1.32	1.10
% $\Delta \pi_3$	-12.95	-12.98	-13.37	-13.74	% $\Delta \pi_0$	13.75	14.33	14.97	15.49
% $\Delta \pi/R$	-1.36	-1.42	-1.45	-1.50	% $\Delta \pi/R$	1.15	1.18	1.19	1.20
% $\Delta CS/R$	2.19	2.49	2.62	2.86	% $\Delta CS/R$	-1.52	-1.70	-1.76	-1.82
% $\Delta W/R$	0.83	1.07	1.17	1.36	% $\Delta W/R$	-0.37	-0.52	-0.57	-0.62

6 Short summary

- Smith builds a structural supermarket-demand model with discrete store choice, continuous expenditure choice, and multi-store Nash pricing.
- He uses store characteristics, a very large household survey, and observed gross profit margins to estimate the model even though direct price data are not available at the relevant level.
- The core economic object is the within-chain cross-effect between stores: if the same chain owns multiple nearby stores, it internalizes substitution across them and raises prices.
- The counterfactuals imply that a complete demerger reduces prices and raises consumer surplus, while mergers among the large chains increase prices and reduce welfare.
- The broad lesson is that multi-store ownership itself is an important source of market power in spatial retail competition.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that multiple-store supermarket chains in the U.K. derive meaningful market power from internalizing substitution across their own outlets. This is not a trivial accounting result: it survives a full structural demand-and-pricing exercise and produces economically sizable price effects in merger and demerger counterfactuals.

7.2 Economic intuition

If two stores belong to different firms, each store cuts price to steal customers from the other. If the two stores belong to the same chain, the firm understands that a price cut at one store partly cannibalizes sales at the other. This softens pricing incentives and raises equilibrium markups. The effect is strongest where the firm's local overlap is greatest and where concentration is already high.

7.3 Takeaway

This paper is a classic example of how structural IO can answer policy questions even when the ideal price data are missing. By combining household shopping choices with firm-level margins, Smith shows that ownership structure alone can materially change supermarket prices. For competition policy, the key message is simple: in spatial retail markets, who owns the stores matters just as much as how many stores there are.